

Geraud Nangue Tasse

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My life philosophy *"What I cannot create, I do not understand" - Richard Feynman*

Education

Ph.D. Computer Science, University of the Witwatersrand , 2021-Present.

M.Sc. Computer Science (with distinction), University of the Witwatersrand, 2019.

B.Sc. Hons Computer Science Mathematics (with distinction), Rhodes University, 2018.

B.Sc. Computer Science, Pure Maths, Applied Maths Physics (distinction), Rhodes Univ., 2015-2017.

Summer Schools

CIFAR Deep Learning + Reinforcement Learning Summer School (*Remotely*), 2020

IBRO-SIMONS Computational Neuroscience Imbizo (*Cape Town, ZA*), 2020

Employment

Research Assistant at RAIL Lab, Remotely , 2020.

I worked on extending my MSc work and writing a journal paper on it.

Tutor, Rhodes University, Grahamstown, ZA, 2015 - 2018.

I was an undergraduate computer science 1 and physics 1 tutor; I also graded computer science practicals.

Awards

Postgraduate PhD Merit Award for MSc with distinction

Deep Learning Indaba-X (ZA-2019) Hackathon Prize

Deep Learning Indaba 2019 Research Poster Prize

Rhodes-Investec Top 100 Award for Academic Excellence

Dean's list for Academic Merit BSc

Open Box Prize for Computer Science Honours

Janinne Franke Prize for the Best Computer Science Honours Project

Programming

Python, C#, Java

Research

Ph.D Computer Science, University of the Witwatersrand, 2021 - Present (ongoing).

Keywords: Reinforcement Learning, Lifelong Learning, Composition. Supervised by Prof Benjamin Rosman

*Title: **Towards Lifelong Reinforcement Learning through Temporal Logics and Zero-Shot Composition.***

M.Sc Computer Science (with distinction), University of the Witwatersrand, 2020.

Keywords: Reinforcement Learning, Task composition, Python. Supervised by Steven James, Prof Benjamin Rosman

*Title: **A Boolean Task Algebra For Reinforcement Learning.***

This work addresses the problem of zero-shot logical composition by first formally defining the composition of tasks as operators acting on a set of tasks in an algebraic structure. This provides a structured way of doing task compositions and a theoretically rigorous way of studying them. We propose a framework for defining lattice algebras and Boolean algebras in particular over the space of tasks. This allows us to formulate new tasks in terms of the negation, disjunction, and conjunction of a set of base tasks. We then show that by learning a new type of goal-oriented value functions and restricting the rewards of the tasks, an agent can solve composite tasks with no further learning.

B.Sc Hons Computer Science Mathematics (with distinction), Rhodes Univ., 2018.

Keywords: Neural Networks, Hyperparameter Optimization, Python. Supervised by James Connan

*Title: **Hypersearch: A Parallel Training Approach For Optimizing Neural Networks Performance.***

This work proposes a novel embarrassingly parallel algorithm, Hypersearch, that takes advantage of both worlds (hyperparameter optimization and parallel programming). Hypersearch works by training multiple neural networks of same architecture but different hyperparameters in parallel while optimizing both networks and hyperparameters. We manage to provide good evidence that Hypersearch not only performs well in realistic settings, but is also competitive against state-of-the-art Hyperparameter optimization alternatives.

Peer-reviewed papers

A Boolean Task Algebra for Reinforcement Learning.

Geraud Nangue Tasse, Steven James and Benjamin Rosman.

In Advances in Neural Information Processing Systems (NeurIPS), 2020. [URL]

A Boolean Task Algebra for Reinforcement Learning.

Geraud Nangue Tasse, Steven James and Benjamin Rosman.

In Beyond "Tabula Rasa" in Reinforcement Learning (BeTR—RL): Agents that remember, adapt, and generalize. Workshop at the International conference on Learning Representations (ICLR), 2020. [URL]

Logical composition for Reinforcement Learning.

Geraud Nangue Tasse, Steven James and Benjamin Rosman.

In 4th Life-long Learning Workshop at the International Conf. on Machine Learning (ICML), 2020. [URL]

Hypersearch: A Parallel Training Approach For Improving Neural Networks Performance.

Geraud Nangue Tasse and James Connan.

Black in AI workshop at the 33rd Conference on Neural Information Processing Systems (NeurIPS), 2019.

Blogs

Blog post on my Boolean task algebra NeurIPS paper.

<https://www.raillab.org/posts/a-boolean-task-algebra-for-reinforcement-learning>, 2019.

Reviewing

2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)

Black in AI workshop at the 34rd Conference on Neural Information Processing Systems (NeurIPS)